This tutorial aims to walk you through the basics of analysis with INSPEX. We will perform a basic single instrument analysis. We will be using the 9 October 2021 event as an example in this tutorial, and analysing the EPD data for the event using the GUI method. For the example, we have run the code on Windows, and recommend you do the same.

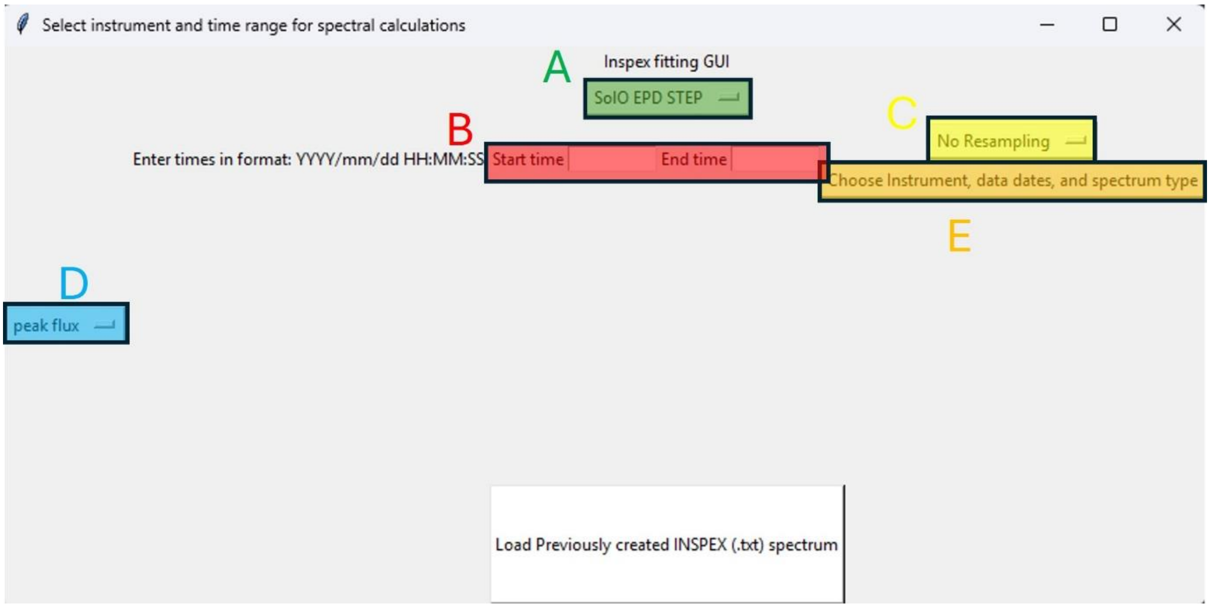
Before we begin, if you are not experienced with python, we will go through the basics of setting up INSPEX. First, from the INSPEX github (https://github.com/SamuelCarter42/INSPEX), download the main branch on to your machine, and extract the files. You can move the folder to your own files, but don't move the contents of the INSPEX main folder around inside it. Next, make sure you have Anaconda installed on your machine. Anaconda Navigator opens as a graphical launcher — find Spyder in the list of applications and click Launch. You should have a console in the Spyder window. Here, use "pip install" to install the package dependencies listed in the README, eg "pip install lmfit".

Step 1: Initialise INSPEX

After installing INSPEX, open and run inspex.py. This sets up the INSPEX environment, defining the functions that will be required for the rest of the process.

Step 2: Defining the time range and spectral type

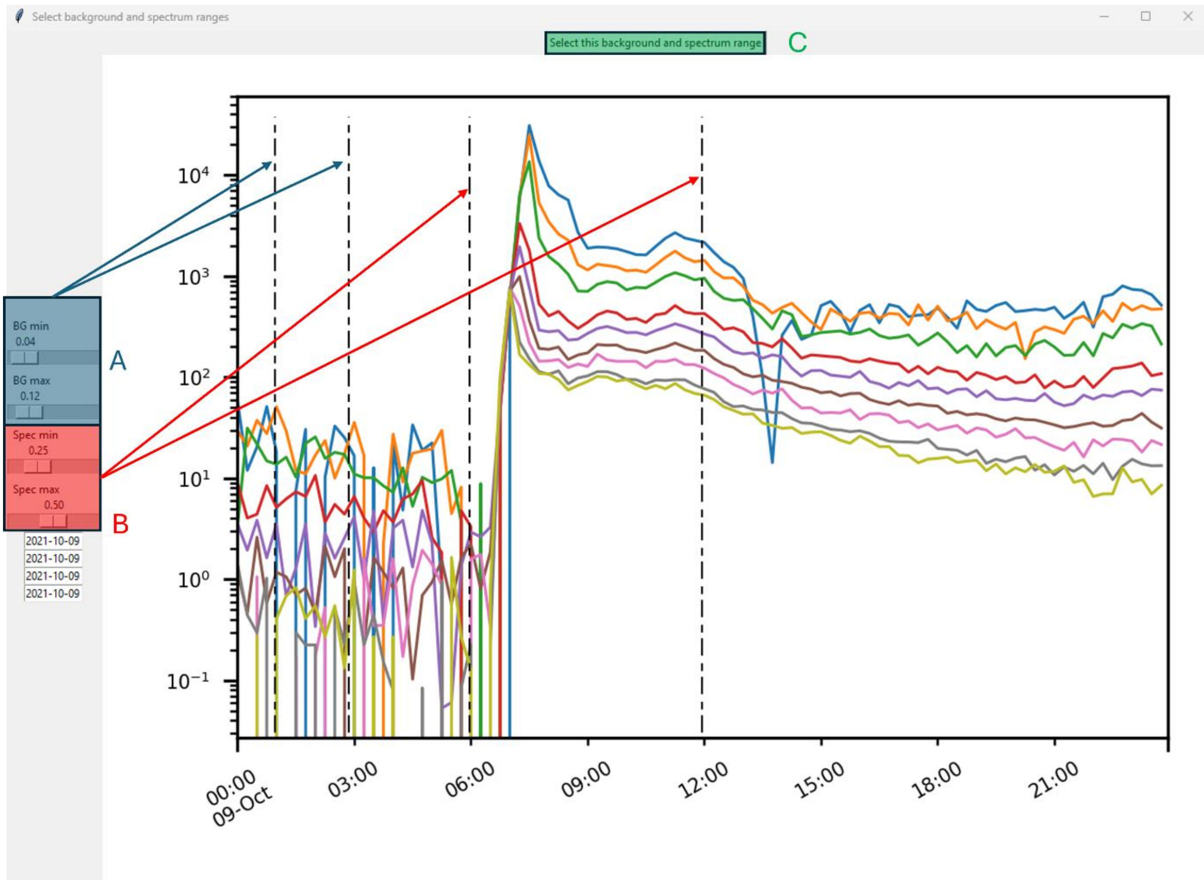
To begin analysis, run instrument_choice() at the command line. This brings up a window from which you may choose the instrument data, date range of interest, spectrum type and resampling for the analysis. We show this in figure 1.



The GUI section marked "A" allows you to select the instrument of interest from a drop-down menu. Click here and select "SolO EPD STEP". Next, look at the section marked "B". Here we enter the time range of interest. We want to load the data for the day of 9 October 2021, so enter "2021/10/09 00:00:00" as the start time and "2021/10/09 23:59:59" as the end time. We want to use a resampled data set for this event, to improve the counting statistics on our data. In section "C", there is a drop-down menu, where we can select "5 minute resampling". We now need to select what sort of spectrum we are generating. In section "D", open the drop-down and select "Peak Flux". Now we are loading the STEP data for 9 October 2021, with 5 minute resampling, and generating a peak flux spectrum. To proceed to the next stage, click the button in section "E", "Choose Instrument, data dates, and spectrum type"

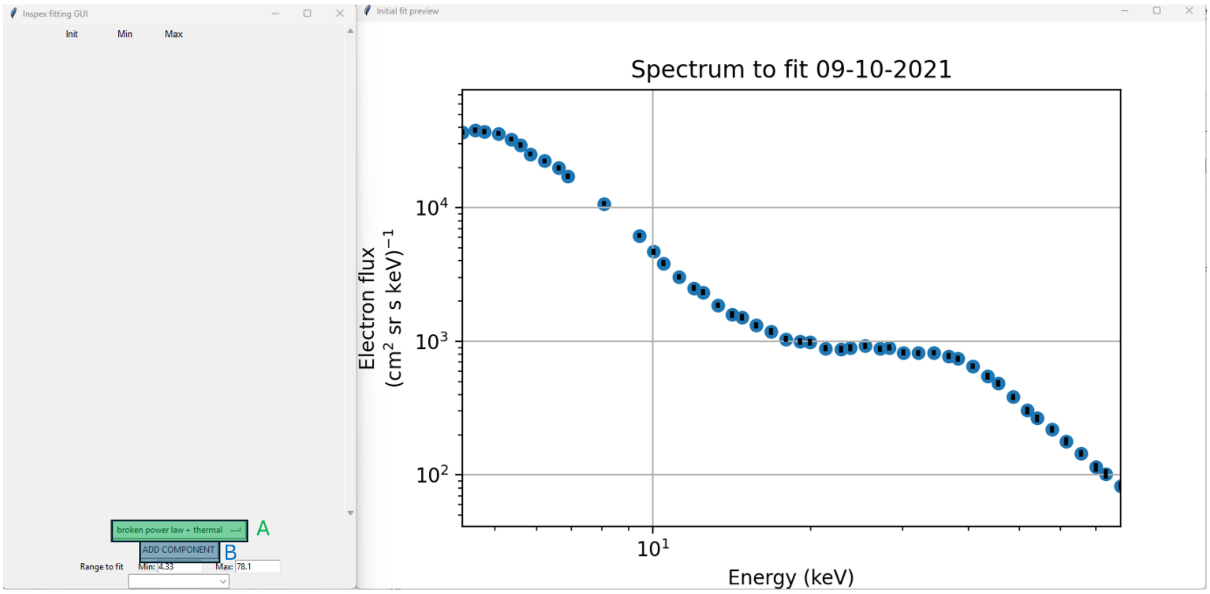
Step 3: Spectrum generation

In this step, we select the data range over which we will generate the spectrum and the range which will act as our background. We have two options for setting these ranges: sliders or entry boxes. These are shown on the window in figure 2. For this tutorial we will use the sliders. When the sliders are moved, the corresponding markers are moved on the time series plot, to indicate the selected times. Background is set with the sliders marked "A". Set the background start to early in the morning of the event, at around 1:00, and the end of the background about 2 hours later, at 3:00. The spectrum generation period is set with the sliders marked "B". Set the spectrum start to just before the event begins, at around 6:00. Set the spectrum end to some point after the event begins to decay away, at around 12:00. Be careful that the background and integration periods do not overlap, and that they start before they end. When this is set, select "Select this background and spectrum range" marked "C".

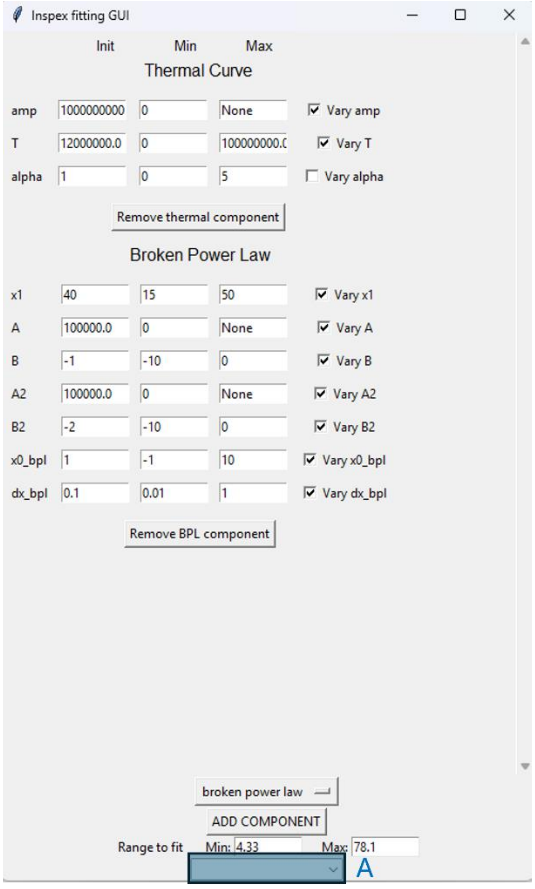


Step 4: Spectral fitting

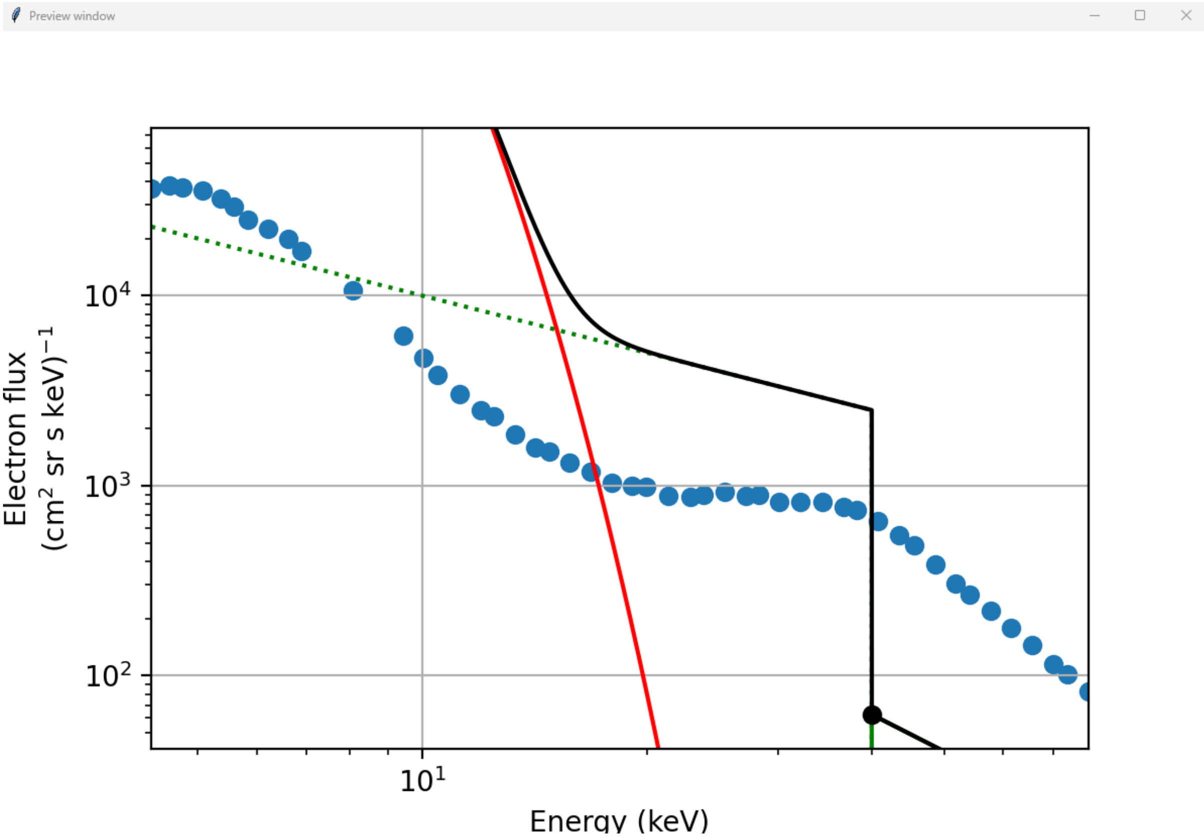
Next, three windows will open so we can begin the fitting process. There is a preview window showing the generated spectrum, and a mostly empty window where we will assemble the model to fit the spectrum we have just generated. There is also a third blank window, which we should minimise as it is there to handle some background tasks. This is normal and can be ignored. For this example, we will fit a single thermal curve at low energy, plus a broken power law at high energy. Figure 3 shows the preview window and the empty GUI. Clicking on the button in section “A” opens a drop-down menu, where you should click on the “thermal” option. Then select “ADD COMPONENT” in section “B”. This will add the parameters for the thermal curve to the fitting GUI, as shown in figure 4. Now, go back into the drop-down menu with the function options (figure 3 section “A”), select the “broken power law” option, and add it to the GUI using the “ADD COMPONENT” button (figure 3 section “B”).



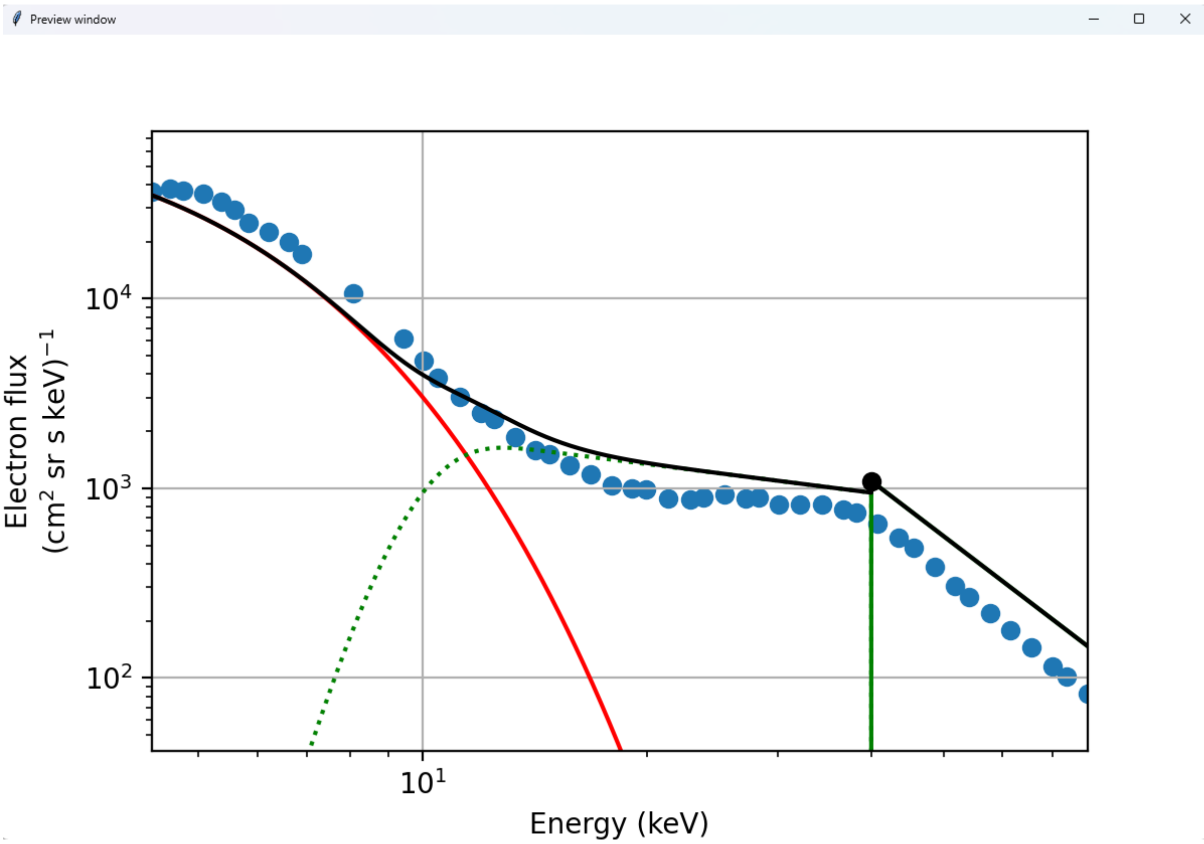
Your fitting GUI window should now look just like the one in figure 4.



For each parameter of the functions, we have 3 entry boxes to allow us to set up the initial guess, minimum and maximum, and a check box to toggle whether each parameter is free to vary. The initial settings are good for this example, but let's preview them to check. Go to the drop-down menu, marked "A" in figure 4, and select "Preview Parameters". This gives us the preview shown in figure 5.

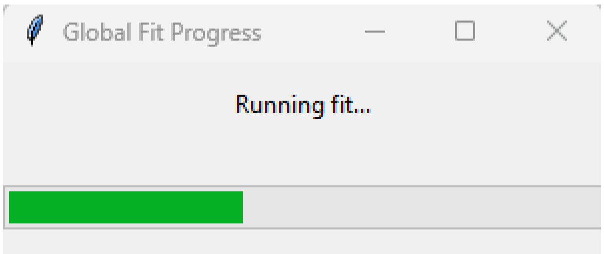


This is not a great starting point for the fit. If we reduce the amplitude parameters of the thermal component and first broken power law component, increase the amplitude on the second broken power law component, and move the broken power law's low energy cut off up, this will fit a lot better. In the GUI window, let's start from the top and work down the parameters. In the initial guess box (leftmost in the row) for thermal curve parameter "amp", we can reduce the value to 100000. We are also going to increase the temperature initial guess to 20000000. Looking further down the GUI we can adjust the "A" parameter of the broken power law component to 6000, "B" to -0.5, "A2" to 70000000, "B2" to -3, "x0_bpl" to 10 and "dx_bpl" to 2. The adjustments to the "x0_bpl" and "dx_bpl" make the most significant change, bringing the low energy cut off up, and giving the drop off a finite width. We can see the result of the changes by selecting the preview parameters option once again. This is shown in figure 6, which looks a lot better. Remember, each spectrum is going to be different, so these starting parameters may not be good for other events.



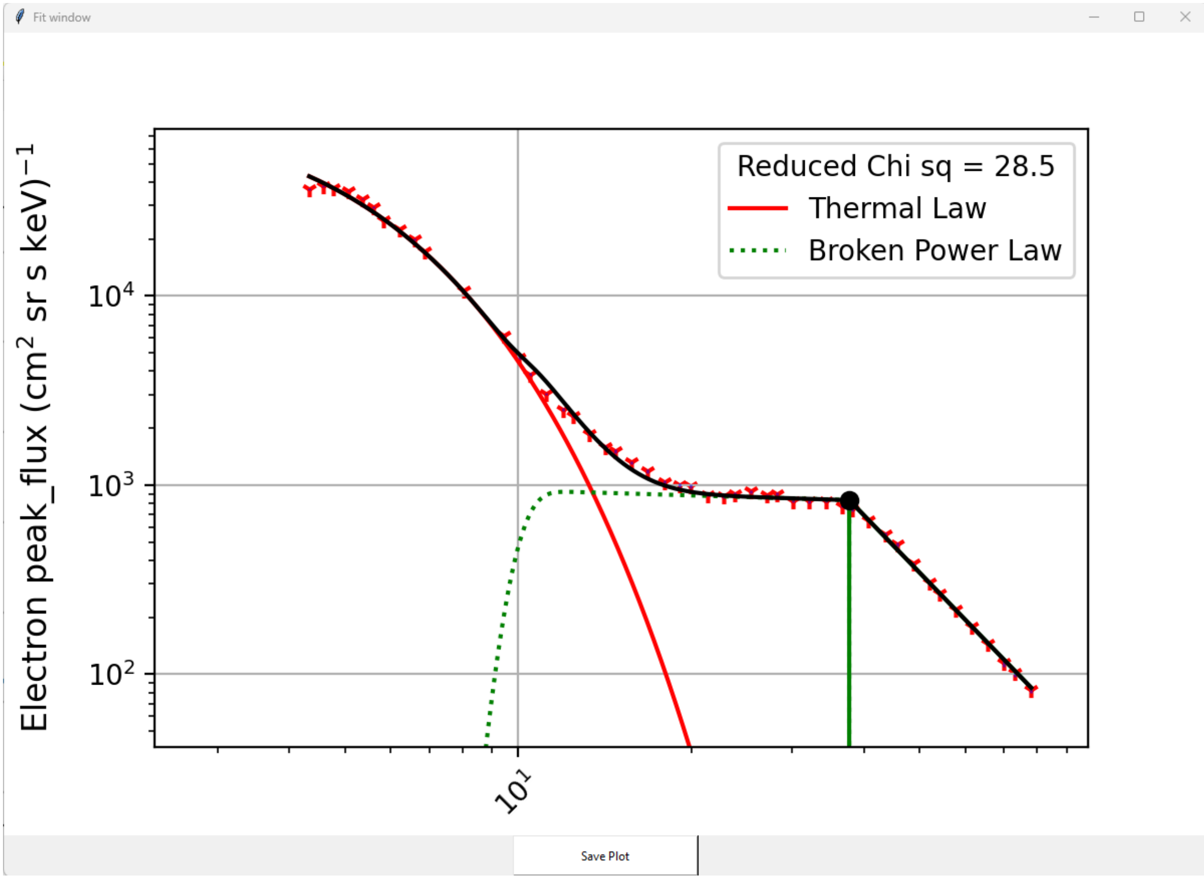
This is a decent approximation of the curve. It doesn't have to be perfect, as this is simply the starting point for the minimiser to work with.

We can now fit our curve! Go to the drop-down menu, marked "A" in figure 4, and select "Perform Fit". While the fit is running, we see the progress bar below in figure 7:



The resulting fit is displayed overlayed on the spectrum, with the reduced chi squared in the legend, as shown in figure 8. Our spectrum should now fit nicely! However, if it doesn't, don't worry! We can make adjustments to the fitted parameters and use preview to see what improves the fit and what doesn't. We can then run the fit again to see if we have improved things. This is an iterative process, so don't feel discouraged if we don't arrive at the best fit first time.

Step 5: Results



At the bottom of the window is a button that allows you to save this plot, if you so wish. Now, in the main GUI window, we have the fitted parameters which have overwritten the initial guesses. We want to save this for later, so go again to the drop-down menu in figure 4. Here the “Save Spectrum” option allows us to save the actual spectrum points we are fitting to, so that in future sessions we can skip the generation process. It saves the x-y coordinates of the flux data points, along with their uncertainty, in a text file. The “Save Parameters” button allows us to save the model we just fitted, with its fitted parameters, so we can load that in future sessions too. These are also saved in a text file, along a string showing which functions we selected for our model. You've now completed a basic INSPEX analysis!